

8th Grade Unit 2 Assessment Guide

General Information	With the introduction of irrational numbers, students now have a full working knowledge of the set of real numbers. The benchmark MA.8.NSO.1.2 builds on MA.6.NSO.1.1. Teachers who find students who are struggling with MA.8.NSO.1.2 may want to support students with Grade 6, Unit 4. If students struggle with the various forms of rational numbers and need further support than what was in Grade 8 Unit 1, teachers may wish to support students using Grade 7, Unit 2, Lesson 3. Students who struggle with which values have perfect squares or perfect cubes may need further support by connecting to visual models. Although memorization is one possible method of knowing these values, it is more helpful if students can generate a list whenever needed. The unit purposefully integrated student understanding of rational and irrational numbers and principal square roots to the solutions of equations in the form of $x^2 = p$. Students often have the common misconception that there is only one solution to equations in the form of $x^2 = p$. This misconception can continue throughout their mathematical journey. Teachers are encouraged to support a student's mathematical journey by ensuring understanding of the solutions of the quadratic in the form of $x^2 = p$ will always have two solutions when the value of p is greater than 0. If students wonder what happens when p is zero or is a negative number, allow them to explore these without incorrectly telling a student that the solution to $x^2 = p$ when $p < 0$ is no solution. Students will be confused when they are later taught that these do have solutions that are imaginary numbers in future courses.
Calculator Information	The questions on the unit assessment are no calculator.
Total Recommended Points	The total recommended points in this guide are 100 points. Teachers can determine how to use the extra 2 points or round the value to 100.

Question	1a-c	Benchmark(s)	MA.8.NSO.1.1; MA.8.NSO.1.2		
a. $2\pi - 9$	<table><tr><td>Point</td><td>C</td></tr></table>	Point	C	Recommended Scoring (18 Total Points)	Consider giving students 6 points for each correct answer. No partial credit should be considered except as described in the additional notes below.
	Point	C			
b. $\sqrt{3}$	<table><tr><td>Point</td><td>T</td></tr></table>	Point	T	Additional Notes	Consider giving students partial credit if the answers to a, b, and c indicate that the student did not note that the scale was 0.5. As a student could be consistent in showing knowledge of position yet ignore the importance of the scale, this should be considered a major error. Consider a deduction of 3 points from each of the answers in this case.
Point	T				
c. $\sqrt[3]{9} - 8$	<table><tr><td>Point</td><td>L</td></tr></table>	Point	L		
Point	L				

Question	2	Benchmark(s)	MA.8.NSO.1.2																																	
	$-\sqrt[3]{60}, -3.2, \frac{27}{4}, \sqrt{48}, 7.05$	Recommended Scoring (12 Total Points)	Consider giving students 6 points for the correct order of the negative values and 6 points for the correct order of the positive values.																																	
Question	3a-c	Benchmark(s)	MA.8.AR.2.3																																	
	<table><tr><th colspan="6">Solutions</th></tr><tr><th></th><th>$x = -4$</th><th>$x = 4$</th><th>$x = -8$</th><th>$x = 8$</th><th>not a real solution</th></tr><tr><td>a.</td><td>$x^2 = 64$</td><td>☐</td><td>☐</td><td>☒</td><td>☒</td><td>☐</td></tr><tr><td>b.</td><td>$x^3 = -64$</td><td>☒</td><td>☐</td><td>☐</td><td>☐</td><td>☐</td></tr><tr><td>c.</td><td>$x^2 = -64$</td><td>☐</td><td>☐</td><td>☐</td><td>☐</td><td>☒</td></tr></table>	Solutions							$x = -4$	$x = 4$	$x = -8$	$x = 8$	not a real solution	a.	$x^2 = 64$	☐	☐	☒	☒	☐	b.	$x^3 = -64$	☒	☐	☐	☐	☐	c.	$x^2 = -64$	☐	☐	☐	☐	☒	Recommended Scoring (18 Total Points)	Consider giving students 6 points for each row. Please consider not giving partial credit for 3a. The student should give both answers for the solutions to this equation.
Solutions																																				
	$x = -4$	$x = 4$	$x = -8$	$x = 8$	not a real solution																															
a.	$x^2 = 64$	☐	☐	☒	☒	☐																														
b.	$x^3 = -64$	☒	☐	☐	☐	☐																														
c.	$x^2 = -64$	☐	☐	☐	☐	☒																														

Question	4a-d	Benchmark(s)	MA.8.NSO.1.2
a.	$8 - \pi$ ☐ < ☐ > ☐ = $2(2.6)$	Recommended Scoring (8 Total Points)	Consider giving 2 points for each correct statement. No partial credit is suggested.
b.	$\frac{9}{4}$ ☐ < ☐ > ☐ = $\frac{3}{4} \cdot \sqrt[3]{8}$		
c.	$(4)^2$ ☐ < ☐ > ☐ = $\sqrt{256}$		
d.	$\sqrt{78}$ ☐ < ☐ > ☐ = 3π		

Question	5	Benchmark(s)	MA.8.AR.2.3			
When x is any real number other than 0 and it is multiplied by itself, the result will ☐ always ☐ sometimes be a ☐ positive ☐ negative value. Therefore, equations in the form $x^2 = p$, where p is a whole number will have ☐ only one solution. ☐ two solutions. ☐ three solutions.	Recommended Scoring (6 Total Points)	No partial credit is suggested. Students should be expected to complete the entire statement with the only correct answers to demonstrate understanding. Any incorrect choices could be the result of guessing instead of understanding the full statement.				
Question	6	Benchmark(s)	MA.8.NSO.1.1			
<table><tr><th>Rational Number</th><th>Irrational Number</th></tr><tr><td> $-0.1\sqrt{144},$ $\frac{57}{10}$ </td><td> $\frac{\sqrt{12}}{2}, -\sqrt[3]{9}, \sqrt{43}, -0.23\pi$ </td></tr></table>	Rational Number	Irrational Number	$-0.1\sqrt{144},$ $\frac{57}{10}$	$\frac{\sqrt{12}}{2}, -\sqrt[3]{9}, \sqrt{43}, -0.23\pi$	Recommended Scoring (36 Total Points)	Consider giving students 6 points for each correctly placed number.
Rational Number	Irrational Number					
$-0.1\sqrt{144},$ $\frac{57}{10}$	$\frac{\sqrt{12}}{2}, -\sqrt[3]{9}, \sqrt{43}, -0.23\pi$					